EXPERIMENT - 14

RAW Sockets to implement Packet sniffing.

AIM:

To write a code using RAW sockets to implement Packet sniffing.

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def Packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        Protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

    #Determine the Protocol.
        Protocol name = ""
        if Protocol == 1:
            Protocol_name = "ICMP"
        elif Protocol == 6:
            Protocol_name = "ICMP"
        elif Protocol == 17:
            Protocol_name = "UDP"
        else:
            Protocol_name = "Unknown Protocol"
```

```
#Print packet details
    Print(f"Protocol: {protocol_name}")
    Print(f"Source Ip: {src_ip}")
    Print(f"Destination Ip: {dst_ip}")
    Print("-"*50)

#Capture Packets on the default network interface.
sniff(iface="wi-fi", prn=packet_callback, filter="ip",
                                        store=0)
```

RESULT:

Thus we have obtained to write a code using RAW sockets to implement Packet sniffing.